Reliable Nuclei Segmentation Across Tissue Types

Reliable instance segmentation and classification
across diverse tissues in PanNuke

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Project 5

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Project summary

Can one model segment and classify nuclei consistently across different tissues and nucleus classes? Following the original Project 5 guidance, we reproduce strong baselines, examine errors by tissue and class, and test targeted improvements. Dead nuclei, including apoptotic and necrotic nuclei, provide a difficult rare-class case. Their detection is one part of the broader question of consistent performance. A completed audit finds persistent tissue and class gaps, while sparse, concentrated Dead labels limit claims of broad coverage [11].

The study examines both training supervision and instance representation. The completed supervision comparison fails both its predefined original-image quality floors and its fixed-window mechanism criterion. The proposed PM-Net study replaces horizontal/vertical distance maps and watershed decoding with point-conditioned queries that predict individual masks and cell types. Matched comparisons will assess detection gains against segmentation quality and a twofold inference-latency limit.

Intellectual merit

By varying training targets and decoder design separately under a shared encoder and evaluation protocol, this study will assess their contributions to rare-nucleus detection and mask quality. The completed audit establishes the need to report tissue and class differences. Planned ablations will test whether a new model improves these failures and whether other classes lose accuracy. Average scores alone cannot establish consistent performance or transfer to unseen tissues.

Broader impacts

Reliable nucleus counts and cell labels support quantitative studies of tissue composition and the tumor microenvironment. Shared evaluation code and tissue-specific error analyses will make improvements in rare-class detection easier to assess alongside segmentation quality and computational cost.

1. Research aims, context, and preliminary evidence

The original guidance asks whether one model can segment and classify nuclei consistently across tissues and cell classes. PanNuke provides 7,901 patches, 19 tissue types and five cell classes [1]. Appearance, density and touching boundaries make instance detection and typing difficult.

Specific aims

Aim 1. Characterize tissue-by-class failures using reproduced HoVer-Net and CellViT-UNI baselines. **Aim 2.** Test supervision for truncated nuclei; the completed comparison fails its predefined criteria. **Aim 3.** Test whether point-conditioned instance prediction improves detection while preserving five-class quality and meeting a twofold inference-cost limit.

Prior work and architectural motivation

HoVer-Net uses horizontal/vertical (HV) maps to separate nuclei [2]; CellViT combines transformer features with this decoding strategy [3]. Proposed PM-Net draws on point-conditioned masks [4], set prediction [5] and spatial adaptation [6]. A matched UNI encoder [10], training schedule and parameter-matched controls will help separate decoder effects from added capacity.

Completed tissue and class audit

Our original three-split reproductions reach mPQ/bPQ 0.4564/0.6613 for HoVer-Net and 0.4995/0.6653 for CellViT-UNI. A new descriptive audit uses nine frozen split/seed pairs of the later x1+M/S baseline [11]. Means average three seeds within each split, then the three splits.

Tissue	mPQ	Connective PQ	Unique test images
Lung	0.4064	0.2542	131
Skin	0.4264	0.3011	100
Colon	0.4370	0.3967	972
Breast	0.5155	0.4398	1,602

Lung, Skin and Colon fall in the bottom five in all three split-level seed means. Differences persist within the same class; the connective groups above each have at least 25 positive images per underlying test fold. This supports a reliability gap, while size, morphology, staining and annotation differences remain possible explanations. It does not establish unseen-tissue transfer as the cause.

Dead detection: justified target, limited coverage

Across all three unique folds, 2,908 Dead nuclei occur in 232 positive images; Lung contains 59.5%, and seven tissues have no Dead labels. In the matched baseline, 49.1% of Dead GT nuclei match a prediction at the IoU criterion, while 74.5% of matched Dead are typed correctly. Detection remains a substantial deficit relative to available annotations.

Frozen EF-P2 changes Dead PQ by +0.0208 in Lung, +0.0093 in Colon and -0.0014 in Stomach. Its overall +0.0070 gain at about 6.1 times inference cost remains mechanism evidence [7,11]. We retain targeted failure analysis and require tissue/class reporting; PM-Net remains a hypothesis. This exploratory audit reuses historical test results, not independent confirmation. Splits 1/2 share test fold 3; unique test counts include folds 1/3 once. No new training or inference was run.

2. Aim 2: supervision for truncated nuclei

Spatial cropping truncates nuclei at training-image boundaries. For instances smaller than 30 pixels, the existing target generator sets HV values to zero. Aim 2 tests whether retaining these geometric targets for newly truncated fragments improves detection; matched intervention arms use identical images and other labels.

Arm	Crop probability	HV target rule
A0: retrained baseline	0.1	Minimum area 30
A1: crop-only control	0.5	Minimum area 30
A2: selective supervision	0.5	Minimum 1 for newly truncated instances with visible area below 30
A3: small-object control	0.5	Minimum 1 for all instances; targets differ only below area 30

All arms use seeds 19 and 1, 130 epochs, matched initialization and optimization, and final-checkpoint evaluation at native resolution without test-time augmentation (TTA). A1-A3 share the augmentation random sequence. Instance identities before and after cropping distinguish newly truncated objects from those already cut by a patch boundary; areas are measured after geometric transformations [8].

Completed original-image development results

All eight training runs and fold-2 evaluations are complete. Relative to A0, A2 improves official Dead PQ by 0.0222 and 0.0269 for seeds 19 and 1, but reduces mPQ by 0.0091 and 0.0034, and bPQ by 0.0062 and 0.0046. Both seeds exceed the predefined 0.002 quality-loss tolerance; seed 1 also loses 0.0044 strict Dead PQ. The frozen development admission rule therefore fails, and no expansion is warranted [8].

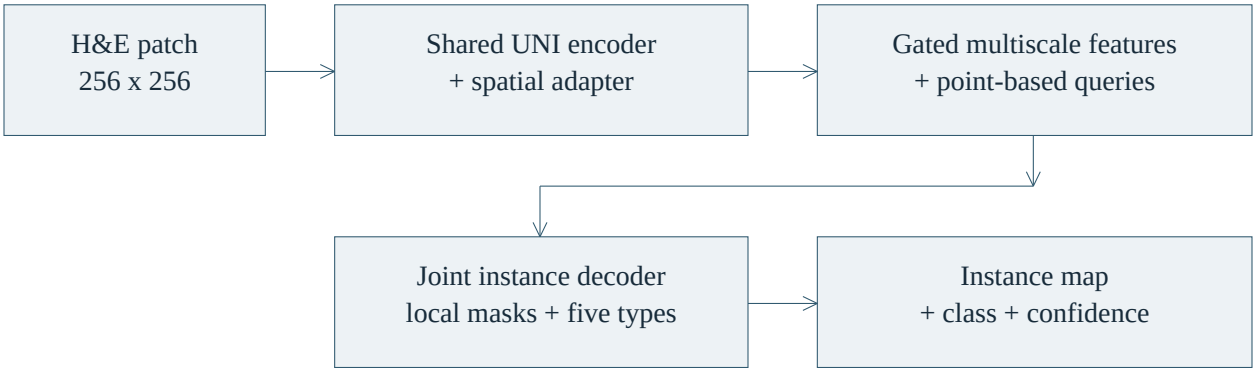
Mechanism analysis

The fixed-window mechanism criterion also fails. Among 116 eligible Dead objects in 38 validation images, A1 and A2 miss 51 and 54 objects for seed 19, and 52 and 51 for seed 1. Mean A1-minus-A2 miss-rate improvement is -0.0086 (paired 95% interval -0.0409 to +0.0177), below the required +0.03 and inconsistent across seeds. Inference covers all 2,523 fixed windows; 77 eligible objects are from Lung, and five only touch the crop edge without losing pixels. Tissue-stratified image bootstrap conditions on these two trained seeds and eligible images. This result does not support the tested selective-HV rescue, nor prove that all visibility-aware approaches fail [8].

Aim 3 tests whether direct mask prediction improves detection of partial or crowded nuclei. PM-Net replaces the HV decoder while retaining a matched encoder and training schedule, complementing Aim 2's controlled changes to supervision.

3. Aim 3: PM-Net, a direct instance model

PM-Net will predict a set of nucleus masks and cell types conditioned on candidate locations [9]. UNI provides tissue context, supplemented by a convolutional adapter for fine spatial features in small and crowded nuclei. The prototype will use native 256 x 256 patches and the same pretrained encoder initialization as CellViT-UNI [10].



Proposed PM-Net. One shared image encoding; no HV maps or watershed.

Architecture and optimization

Shared image features. A compact CNN adapter adds higher-resolution spatial features to UNI tissue representations, with learned gates combining the scales for the point and mask heads.

Point-conditioned queries. A supervised heatmap predicts candidate locations and types; each selected point initializes a feature vector, or query, for one proposed nucleus. Point targets lie inside visible instance masks so that cropped objects retain valid locations. Training-only auditing found a maximum of 276 instances per patch: 256 queries would leave at least 20 instances unrepresentable in one training image. Capacity must exceed that observed maximum with a safety margin; 512 covers the original training patches, with augmented counts and computational cost still to verify.

Joint mask and type prediction. Queries attend to local features and tissue context to predict masks, confidence, and five-class probabilities. One-to-one matching associates predictions with labeled nuclei by location, class, and mask agreement. Training combines point focal loss, mask cross-entropy and Dice losses, and type cross-entropy, with weights selected on validation data. Direct heatmap supervision provides a training signal for the point head despite non-differentiable point selection.

Instance decoding. A confidence rule selected on validation data resolves overlapping masks before mapping them to native image coordinates for scoring. We will assess how minimum-area filtering affects small-instance recall separately from the training-target cutoff in Aim 2.

Controlled ablations

A basic query decoder will be compared with versions that add point initialization and gated fusion separately. A plain adapter with matched parameter count will distinguish architectural effects from added capacity. The initial study retains five-class prediction throughout; truncation supervision and auxiliary type grouping are subsequent extensions rather than prerequisites for the core model.

4. Experimental design and evaluation

Development and confirmation

Development pairs seeds 19 and 1 on training fold 1 and validation fold 2, using the existing 130-epoch schedule and encoder unfreezing at epoch 25. The encoder initialization and training recipe will be matched across PM-Net and CellViT-UNI. Architecture, losses, augmentation, decoding, and decision criteria will be fixed before confirmation.

If development criteria and resources permit, confirmation will use train/validation/test folds 1/2/3, 2/1/3, and 3/2/1, evaluating final checkpoints for every split and seed before averaging. Splits 1 and 2 share test fold 3. Results will disclose prior exploratory use of test folds and the lack of identifiers needed to verify patient-level separation. The official splits assess new images from represented tissues. They do not establish transfer to tissue types excluded from training.

Endpoints and statistical analysis

Official mPQ summarizes type-aware panoptic quality (PQ), while bPQ ignores type. Both can hide subgroup losses. We will report a 19-tissue by five-class PQ heatmap and paired changes, per-tissue mPQ/bPQ, and the spread and lowest tissue scores. Dead recall and precision, matched-instance typing, and merge/split rates will distinguish detection gains from better masks or class labels. Strict scores retain absent-class false-positive penalties.

Comparisons with matched CellViT-UNI and the crop-only control will use paired, tissue-stratified image bootstrap intervals. Crop windows from each source patch will form a single resampling cluster. Each seed will be reported separately, since these intervals measure image-sampling variation for the trained models. Each subgroup will include image and nucleus counts. Absent classes are not applicable; sparse estimates receive intervals and cautious interpretation. Without patient identifiers, image intervals cannot establish patient-level generalization.

PM-Net development criteria

Before training, we propose fixing a primary target of at least 0.03 mean improvement in interior Dead recall, with a positive paired 95% lower confidence bound. Mean Dead PQ must improve by at least 0.005, and both recall and Dead PQ changes must be positive in each seed. Per-seed mPQ/bPQ losses are limited to 0.002; strict mPQ and strict Dead PQ must not decline, and mean merge/split rates must not increase. These criteria apply to PM-Net, independently of the frozen supervision experiment.

End-to-end inference, including decoding, is limited to twice the matched single-scale baseline latency. Repeated measurements on the same hardware and at the same batch settings will quantify latency, peak memory, and variability under shared-device load. A prototype meeting both quality and cost criteria will qualify for cross-split confirmation as resources permit.

5. Milestones, feasibility, and deliverables

Schedule	Milestone
Week 1	Complete the supervision analysis and baseline/error tables (minimum result).
Week 2	Finalize PM-Net design; implement and verify matching, masks, query capacity, and runtime.
Weeks 3-4	Complete the two-seed development comparison and core ablations (target result).
Weeks 5-6	Finalize the report; pursue cross-split and external evaluation if the prototype qualifies and resources permit.

The six-week schedule begins at proposal agreement, with three levels of scope:

Minimum result. Reproduced baselines, the completed supervision comparison, and the completed tissue-by-class audit with PQ, sample support, error rates, and uncertainty. These directly address the original guidance.

Target result. A tested PM-Net prototype and two-seed ablations that establish whether direct instance prediction improves Dead recall and PQ within the quality and latency criteria. A clear account of any remaining tradeoff is also a useful outcome.

Stretch goals. Cross-split confirmation and optional external evaluation with class mapping and overlap exclusion. A separate, small tissue-holdout study could exclude chosen tissues from both training and tuning before testing them. This extension needs a frozen protocol and resources. Pathology pretraining limits claims of entirely unseen tissue exposure.

Feasibility and main risks

Rare labels and annotation uncertainty. Rare tissue-class combinations limit precision. Counts and paired intervals will accompany results, and gains concentrated in one tissue will receive a limited claim. Detector disagreement remains uncertain without expert review.

Localization versus mask quality. In crowded regions, a correctly located query may still produce a mask with insufficient overlap for matching. Separating missed points, low-overlap masks, duplicates, merges, and splits will show whether localization or mask prediction limits performance.

Experimental scope and resources. The encoder and training recipe remain fixed during the decoder ablations. We estimate 2-4 A100 GPU-days for development and a further 4-8 for confirmation, subject to allocation. Available storage and checkpoint requirements will be checked before scheduling; resumable training will reduce losses from interruptions.

Deliverables and broader impact

The target deliverables are PM-Net code and trained models, versioned evaluation configurations, model/data provenance, comparison tables, and a final report with annotated failure cases. Together, these outputs will document how supervision and instance representation affect rare-nucleus detection under matched conditions.

Tissue-specific error reporting will help researchers judge the coverage and computational cost of apparent improvements in cell-composition measurements.

References and project evidence

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Versioned project documentation

[8] BDD26 Project Team. (2026). Visibility-supervision experiment: frozen specification, completed development evaluations, and fixed-window mechanism analysis. Internal research protocol and experiment records, Johns Hopkins University.

Frozen supervision specification

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Versioned new-model concept

[10] Chen, R. J., et al. (2024). Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30, 850-862.

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[11] BDD26 Project Team. (2026). Tissue and class consistency audit of frozen PanNuke evaluations. Nine matched split/seed pairs, unique-fold support counts, and exploratory paired image bootstrap.

Audit methods, results, and reproducible sources

Team contributions to this proposal

This page is separate from the main proposal.

The team's proposal contributions are summarized below.

Zhiyu Song

Helped shape the research question around tissue- and class-specific segmentation reliability, and contributed to the project's biological motivation and scope.

Yuchen Ai

Contributed to literature review and method comparisons, helping frame the baseline choices and the proposed point-conditioned instance-segmentation approach.

Xinhao Jin

Implemented and maintained core segmentation and evaluation code, analyzed tissue-by-class results and failure modes, and integrated findings into the proposal.

Zheng Jian

Contributed to experimental-design ideas, validation and metric choices, and interpretation of results; reviewed the PM-Net ablations and feasibility criteria.